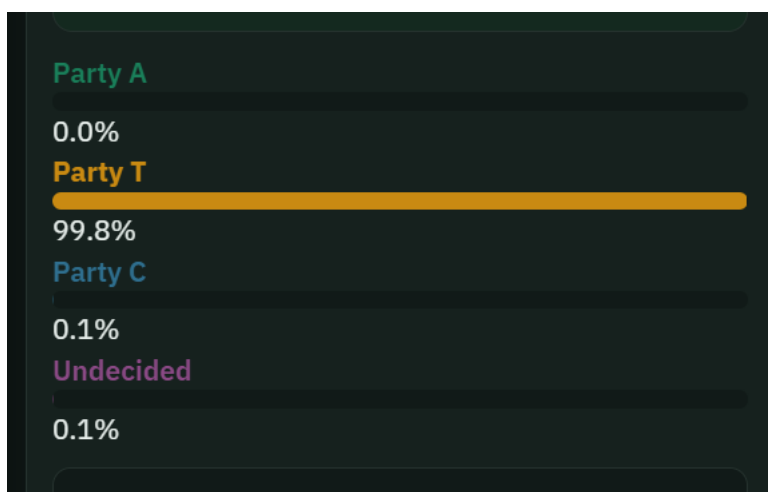
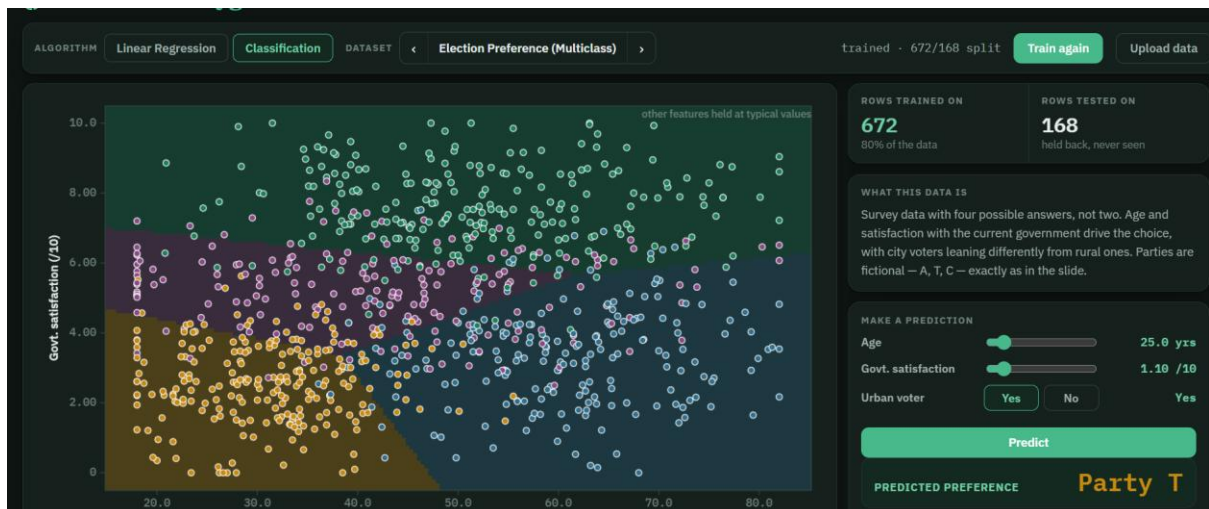


Home Assignment-week1day1

1) Classification Algorithm

a) Age <30 and Govt satisfaction less, Urban
validate the percentage



Party A

0.0%

Party T

99.8%

Party C

0.1%

Undecided

0.1%

Step 1 — each class gets its own score = base + (weight × this voter's value), added up
Party A -15.1 pts ← base points + 0.069×25.0 (Age) = $+1.72 + 2.44 \times 1.10$ (Govt. satisfaction) = $+2.68 - 1.22 \times \text{Yes}$ (Urban voter) = -1.22 ————— = **-12.0 pts**
Party T 7.59 pts ← base points - 0.082×25.0 (Age) = $-2.05 - 1.10 \times 1.10$ (Govt. satisfaction) = $-1.21 + 0.940 \times \text{Yes}$ (Urban voter) = $+0.940$ ————— = **5.27 pts**
Party C -3.10 pts ← base points + 0.145×25.0 (Age) = $+3.63 - 0.514 \times 1.10$ (Govt. satisfaction) = $-0.566 - 1.49 \times \text{Yes}$ (Urban voter) = -1.49 ————— = **-1.53 pts**
Undecided -2.59 pts ← base points + 0.021×25.0 (Age) = $+0.523 + 0.746 \times 1.10$ (Govt. satisfaction) = $+0.821 - 0.543 \times \text{Yes}$ (Urban voter) = -0.543 ————— = **-1.79 pts** ————— **highest score wins → Party T**

Step 2a — convert each score to e^score (always positive) Party A $e^{-12.0} = 0.000$ Party T $e^{5.27} = 194$ Party C $e^{-1.53} = 0.216$ Undecided $e^{-1.79} = 0.168$ —————
Total (add all 4 up) = 195

Step 2b — each class's share = its own e^score ÷ that same total (195) Party A $0.000 \div 195 = 0.0\%$ Party T $194 \div 195 = 99.8\%$ Party C $0.216 \div 195 = 0.1\%$ Undecided $0.168 \div 195 = 0.1\%$ ————— All 4 shares add back up to 100% — that's the whole point of dividing by the same total. e (≈ 2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

The trained model

Party A score = **-15.1 pts** ← base points (starting score) + $0.069 \times \text{Age} + 2.44 \times \text{Govt. satisfaction} - 1.22 \times \text{Urban voter}$
Party T score = **7.59 pts** ← base points (starting score) - $0.082 \times \text{Age} - 1.10 \times \text{Govt. satisfaction} + 0.940 \times \text{Urban voter}$
Party C score = **-3.10 pts** ← base points (starting score) + $0.145 \times \text{Age} - 0.514 \times \text{Govt. satisfaction} - 1.49 \times \text{Urban voter}$
Undecided score = **-2.59 pts** ← base points (starting score) + $0.021 \times \text{Age} + 0.746 \times \text{Govt. satisfaction} - 0.543 \times \text{Urban voter}$

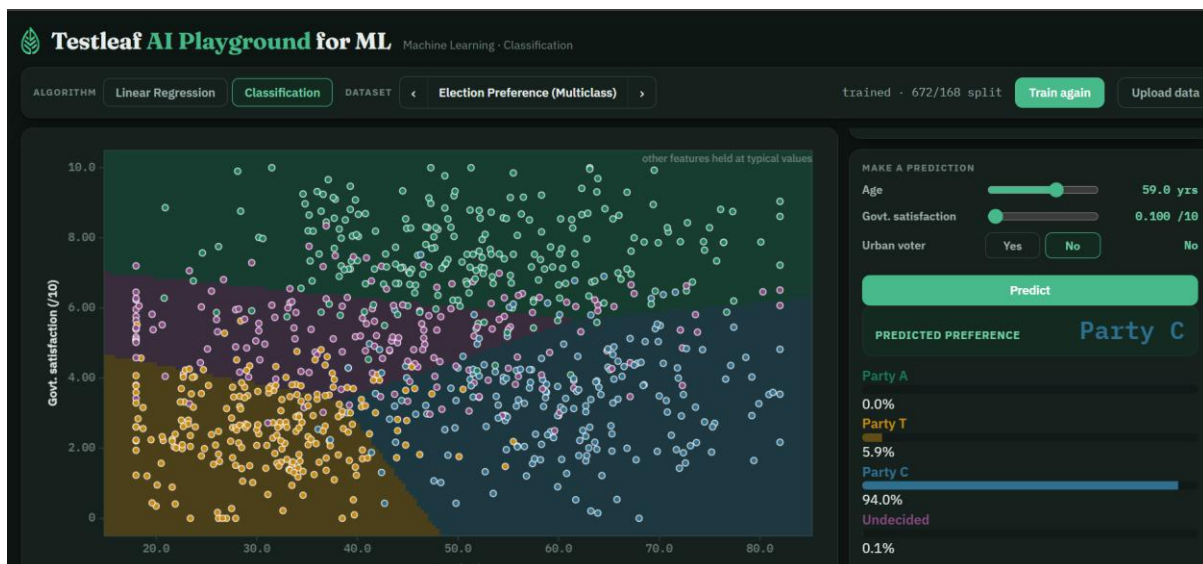
Sample calculation — one real row

45.7 yrs · 6.43 / 10 · not urban voter Party A **3.66** Party T **-3.25** Party C **0.222** Undecided **3.17** ————— **highest wins → Party A** actual Party A

With more than two possible answers, the model keeps **one score per class** — four separate sums from the same inputs. Whichever score comes out highest is the prediction. That is all “multiclass” means.

b) Age >50 and Govt satisfaction less, non- Urban

validate the percentage



Step 1 — each class gets its own score = base + (weight × this voter's value), added up

Party A -15.1 pts ← base points + 0.069×59.0 (Age) = +4.06 + 2.44×0.100 (Govt. satisfaction) = +0.244 – $1.22 \times \text{No}$ (Urban voter) = +0 ————— = -10.8 pts

Party T 7.59 pts ← base points – 0.082×59.0 (Age) = -4.84 – 1.10×0.100 (Govt. satisfaction) = -0.110 + $0.940 \times \text{No}$ (Urban voter) = +0 ————— = 2.64 pts

Party C -3.10 pts ← base points + 0.145×59.0 (Age) = +8.56 – 0.514×0.100 (Govt. satisfaction) = -0.051 – $1.49 \times \text{No}$ (Urban voter) = +0 ————— = 5.41 pts

Undecided -2.59 pts ← base points + 0.021×59.0 (Age) = +1.23 + 0.746×0.100 (Govt. satisfaction) = +0.075 – $0.543 \times \text{No}$ (Urban voter) = +0 ————— = -1.28 pts ————— **highest score wins → Party C**

Step 2a — convert each score to e^{score} (always positive) Party A $e^{-10.8} = 0.000$ Party T $e^{2.64} = 14.0$ Party C $e^{5.41} = 223$ Undecided $e^{-1.28} = 0.278$ ————— **Total (add all 4 up) = 237**

Step 2b — each class's share = its own $e^{\text{score}} \div$ that same total (237) Party A $0.000 \div 237 = 0.0\%$ Party T $14.0 \div 237 = 5.9\%$ Party C $223 \div 237 = 94.0\%$ Undecided $0.278 \div 237 = 0.1\%$ ————— All 4 shares add back up to 100% — that's the whole point of dividing by the same total. e (≈ 2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

The trained model

Party A score = -15.1 pts ← base points (starting score) + $0.069 \times \text{Age}$ + $2.44 \times \text{Govt. satisfaction}$ – $1.22 \times \text{Urban voter}$

Party T score = 7.59 pts ← base points (starting score) – $0.082 \times \text{Age}$ – $1.10 \times \text{Govt. satisfaction}$ + $0.940 \times \text{Urban voter}$

Party C score = -3.10 pts ← base points (starting score) + $0.145 \times \text{Age}$ – $0.514 \times \text{Govt. satisfaction}$ – $1.49 \times \text{Urban voter}$

Undecided score = -2.59 pts ← base points (starting score) + $0.021 \times \text{Age}$ + $0.746 \times \text{Govt. satisfaction}$ – $0.543 \times \text{Urban voter}$

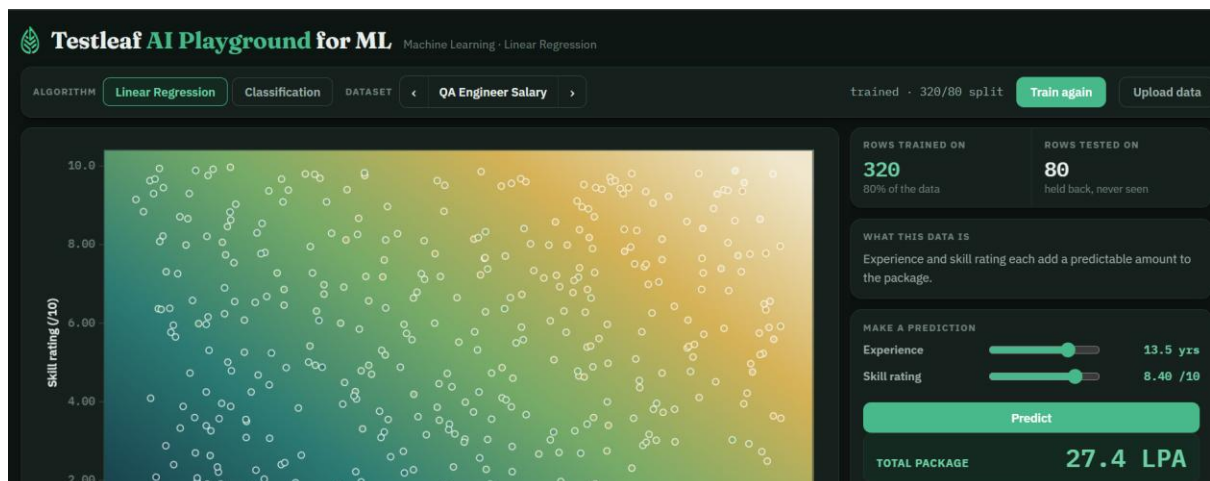
Sample calculation — one real row

45.7 yrs · 6.43 /10 · not urban voter Party A **3.66** Party T **-3.25** Party C **0.222** Undecided **3.17** ————— **highest wins → Party A** actual Party A

With more than two possible answers, the model keeps **one score per class** — four separate sums from the same inputs. Whichever score comes out highest is the prediction. That is all “multiclass” means.

2) Linear regression

a) QA Salary- Check the Calculation for experience and skill



Base salary — the starting value, before any feature counts **3.10 LPA** Experience **0.966 LPA per year × 13.5 yrs = +13.0 LPA** Skill rating **1.34 LPA per point × 8.40 /10 = +11.3 LPA** ————— **TOTAL 27.4 LPA**

The trained model

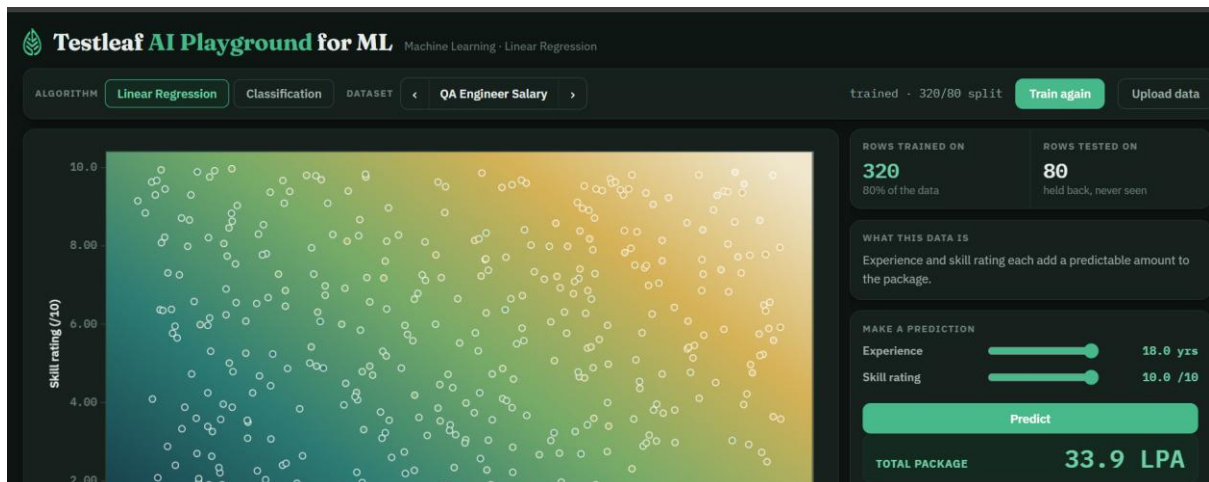
Salary_LPA in LPA = **3.10 LPA** ← base salary (minimum) + **0.966 LPA per year × Experience** + **1.34 LPA per point × Skill rating**

Sample calculation — one real row from the data

9.46 yrs · 5.78 /10 **3.10** (base) + **0.966 × 9.46 = +9.14** + **1.34 × 5.78 = +7.75**
 ————— **model says 20.0 LPA** actual value 19.4 LPA

Read it as a rate card. The first number is the **base salary** — the minimum you start from before any feature is counted. Every line after it says what one more unit is worth. Press Predict to see it added up.

With High experience and High Skill



Base salary — the starting value, before any feature counts 3.10 LPA
 Experience 0.966 LPA per year × 18.0 yrs = +17.4 LPA
 Skill rating 1.34 LPA per point × 10.0 / 10 = +13.4 LPA
 ————— TOTAL 33.9 LPA

The trained model

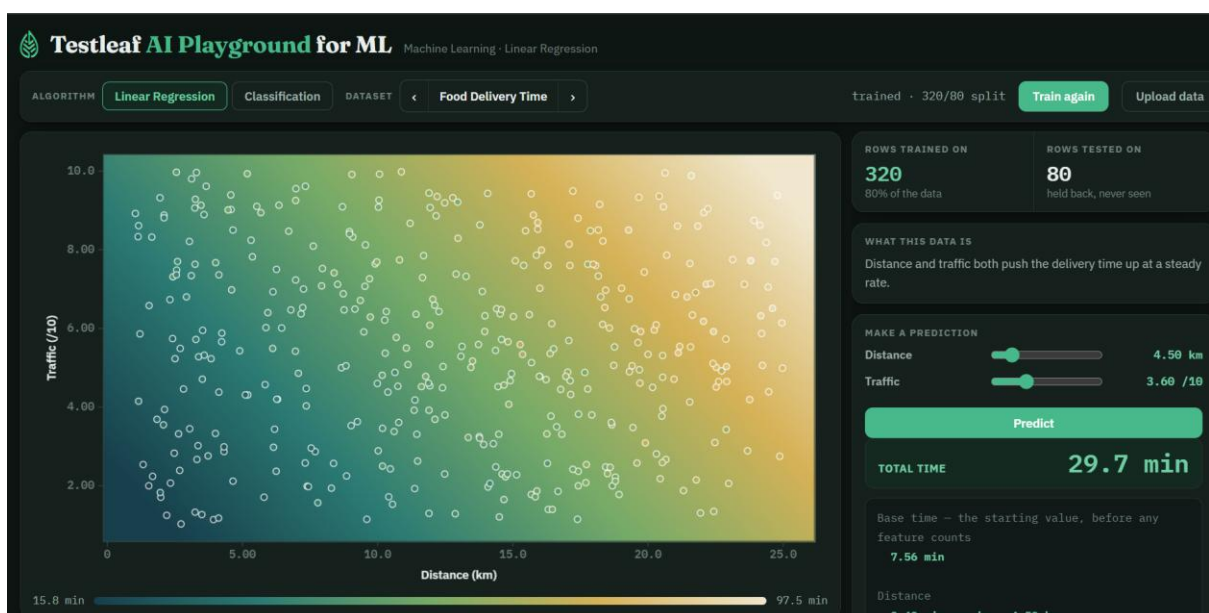
Salary_LPA in LPA = 3.10 LPA ← base salary (minimum) + 0.966 LPA per year × Experience + 1.34 LPA per point × Skill rating

Sample calculation — one real row from the data

9.46 yrs · 5.78 / 10 3.10 (base) + 0.966 × 9.46 = +9.14 + 1.34 × 5.78 = +7.75
 ————— model says 20.0 LPA actual value 19.4 LPA

Read it as a rate card. The first number is the base salary — the minimum you start from before any feature is counted. Every line after it says what one more unit is worth. Press Predict to see it added up.

b) Food Delivery time- Check the Calculation for price and time



Base time — the starting value, before any feature counts **7.56 min** Distance **2.40 min**
per km × **4.50 km** = **+10.8 min** Traffic **3.16 min per point** × **3.60 /10** = **+11.4 min**
————— **TOTAL 29.7 min**

The trained model

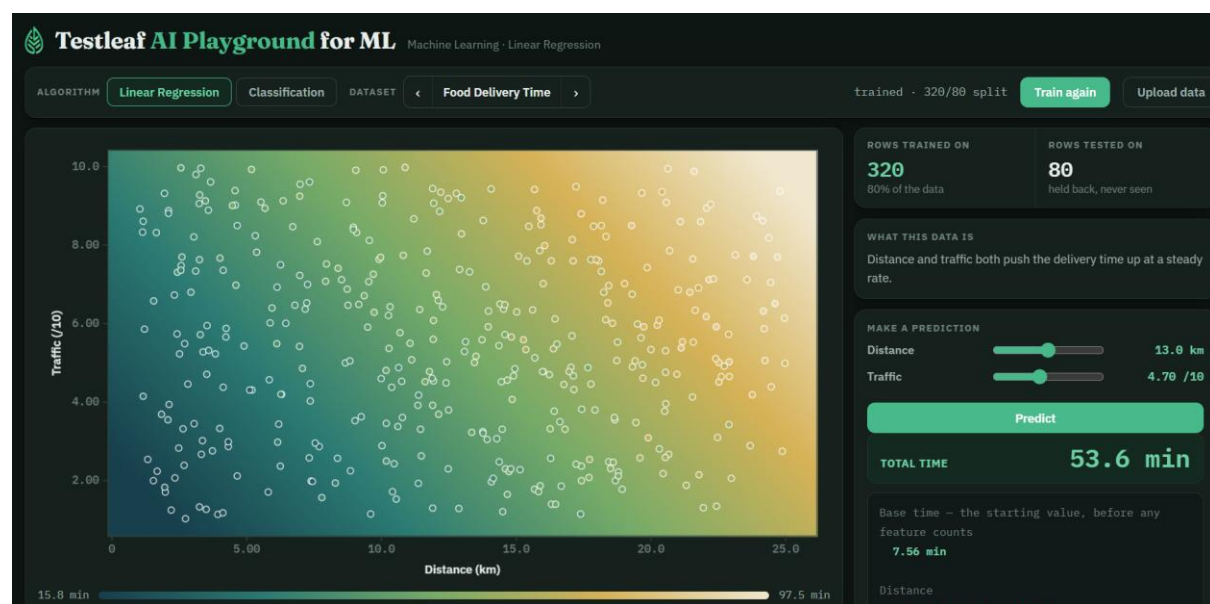
Delivery_Minutes in min = **7.56 min** ← base time (minimum) + **2.40 min per km** ×
Distance + **3.16 min per point** × Traffic

Sample calculation — one real row from the data

8.96 km · 8.46 /10 **7.56** (base) + **2.40 × 8.96** = **+21.5** + **3.16 × 8.46** = **+26.7**
————— **model says 55.8 min** actual value 55.4 min

Read it as a rate card. The first number is the **base time** — the minimum you start from
before any feature is counted. Every line after it says what one more unit is worth. Press
Predict to see it added up.

With Average time and Distance coverage



Base time — the starting value, before any feature counts **7.56 min** Distance **2.40 min**
per km × **13.0 km** = **+31.2 min** Traffic **3.16 min per point** × **4.70 /10** = **+14.8 min**
————— **TOTAL 53.6 min**

The trained model

Delivery_Minutes in min = **7.56 min** ← base time (minimum) + **2.40 min per km** ×
Distance + **3.16 min per point** × Traffic

Sample calculation — one real row from the data

8.96 km · 8.46 /10 **7.56** (base) + **2.40 × 8.96** = **+21.5** + **3.16 × 8.46** = **+26.7**
————— **model says 55.8 min** actual value 55.4 min

Read it as a rate card. The first number is the base time — the minimum you start from before any feature is counted. Every line after it says what one more unit is worth. Press Predict to see it added up.